

✓ Box Star Print

%%writefile pattern1.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    while (i<=n){
        int j=1;
        while (j<=n){
            cout<<"* "<<" ";
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}
```

 Writing pattern1.cpp

```
!g++ pattern1.cpp -o pattern1
!./pattern1
```

 5

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

✓ Box Numeric Printing

%%writefile pattern2.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    while (i<=n){
        int j=1;
        while (j<=n){
            cout<<i<<" ";
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}
```

 Writing pattern2.cpp

```
!g++ pattern2.cpp -o pattern2
!./pattern2
```

 5

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

%%writefile pattern3.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    while (i<=n){
        int j=1;
```

```

        while (j<=n){
            cout<<j<<" ";
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}

```

➡ Writing pattern3.cpp

```

!g++ pattern3.cpp -o pattern3
!./pattern3

```

➡ 5

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

```

%%writefile pattern4.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    while (i<=n){
        int j=1;
        while (j<=n){
            cout<<n-j+1<<" ";
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}

```

➡ Writing pattern4.cpp

```

!g++ pattern4.cpp -o pattern4
!./pattern4

```

➡ 5

```

5 4 3 2 1
5 4 3 2 1
5 4 3 2 1
5 4 3 2 1
5 4 3 2 1

```

%%writefile pattern5.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    int count = 1;
    while (i<=n){
        int j=1;
        while (j<=n){
            cout<<count<<" ";
            count+=1;
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}

```

➡ Writing pattern5.cpp

```

!g++ pattern5.cpp -o pattern5
!./pattern5

```

➡ 5

```

1 2 3 4 5
6 7 8 9 10

```

```
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
```

✓ Triangle Star Printing

```
%%writefile pattern6.cpp
```

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<" "<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern6.cpp

```
!g++ pattern6.cpp -o pattern6
!./pattern6
```

 5

```
*
* *
* * *
* * * *
* * * * *
```

```
%%writefile pattern21.cpp
```

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        // space print
        int space = n-row;
        while (space){
            cout<<" "<<" ";
            space-=1;
        }
        // * print
        int col=1;
        while (col<=row){
            cout<<" "<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern21.cpp

```
!g++ pattern21.cpp -o pattern21
!./pattern21
```

 5

```
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
```

```
%%writefile pattern22.cpp
```

```
#include<iostream>
using namespace std;
```

```
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=n-row+1){
            cout<<" "<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern22.cpp

```
!g++ pattern22.cpp -o pattern22
!./pattern22
```

 5

```
* * * * *
* * * *
* * *
* *
*
*
```

%%writefile pattern23.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        // space print
        int space = row-1;
        while (space){
            cout<<" "<<" ";
            space-=1;
        }
        // * print
        int col=1;
        while (col<=n-row+1){
            cout<<" "<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern23.cpp

```
!g++ pattern23.cpp -o pattern23
!./pattern23
```

 5

```
* * * * *
* * * *
* * *
* *
*
*
```

▼ Triangle Numeric Printing

%%writefile pattern7.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<row<<" ";
        }
    }
}
```

```

        col+=1;
    }
    cout<<endl;
    row+=1;
}
}

```

➡ Writing pattern7.cpp

```

!g++ pattern7.cpp -o pattern7
!./pattern7

```

➡

```

5
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```

%%writefile pattern8.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<col<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

➡ Writing pattern8.cpp

```

!g++ pattern8.cpp -o pattern8
!./pattern8

```

➡

```

5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

%%writefile pattern9.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    int count =1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<count<<" ";
            count+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

➡ Writing pattern9.cpp

```

!g++ pattern9.cpp -o pattern9
!./pattern9

```

➡

```

5
1
2 3
4 5 6
7 8 9 10

```

11 12 13 14 15

Method 1

%%writefile pattern10.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        int value = row;
        while (col<=row){
            cout<<value<<" ";
            value+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern10.cpp

```
!g++ pattern10.cpp -o pattern10
!./pattern10
```

 5
1
2 3
3 4 5
4 5 6 7
5 6 7 8 9

Method 2

%%writefile pattern11.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<row+col-1<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern11.cpp

```
!g++ pattern11.cpp -o pattern11
!./pattern11
```

 5
1
2 3
3 4 5
4 5 6 7
5 6 7 8 9

%%writefile pattern12.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
```

```

        int col=1;
        while (col<=row){
            cout<<row-col+1<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

 Writing pattern12.cpp

```

!g++ pattern12.cpp -o pattern12
!./pattern12

```

 5
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1

▼ Box Alphabet Printing

```
%%writefile pattern13.cpp
```

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    char ch = 'A';
    while (row<=n){
        int col=1;
        while (col<=n){
            cout<<ch<<" ";
            col+=1;
        }
        cout<<endl;
        ch+=1;
        row+=1;
    }
}

```

 Writing pattern13.cpp

```

!g++ pattern13.cpp -o pattern13
!./pattern13

```

 5
A A A A A
B B B B B
C C C C C
D D D D D
E E E E E

```
%%writefile pattern14.cpp
```

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    char ch = 'A';
    int count = 1;
    while (row<=n){
        int col=1;
        while (col<=n){
            cout<<ch<<" ";
            ch+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

➤ Writing pattern14.cpp

```
!g++ pattern14.cpp -o pattern14
!./pattern14
```

➤ 5

```
A B C D E
F G H I J
K L M N O
P Q R S T
U V W X Y
```

%%writefile pattern15.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int i=1;
    while (i<=n){
        int j=1;
        while (j<=n){
            char ch = 'A'+j-1;
            cout<<ch<<" ";
            j+=1;
        }
        cout<<endl;
        i+=1;
    }
}
```

➤ Writing pattern15.cpp

```
!g++ pattern15.cpp -o pattern15
!./pattern15
```

➤ 5

```
A B C D E
A B C D E
A B C D E
A B C D E
A B C D E
```

%%writefile pattern16.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    // char ch = 'A';
    while (row<=n){
        int col=1;
        while (col<=n){
            char ch = 'A'+row+col-2;
            cout<<ch<<" ";
            ch+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

➤ Writing pattern16.cpp

```
!g++ pattern16.cpp -o pattern16
!./pattern16
```

➤ 5

```
A B C D E
B C D E F
C D E F G
D E F G H
E F G H I
```


▼ Triangle Alphabet Printing

%%writefile pattern17.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    char ch='A';
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<ch<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
        ch+=1;
    }
}
```

 Writing pattern17.cpp

```
!g++ pattern17.cpp -o pattern17
!./pattern17
```

 5
A
B B
C C C
D D D D
E E E E E

%%writefile pattern18.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    char ch = 'A';
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<ch<<" ";
            ch+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

 Writing pattern18.cpp

```
!g++ pattern18.cpp -o pattern18
!./pattern18
```

 5
A
B C
D E F
G H I J
K L M N O

%%writefile pattern19.cpp

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        char ch = 'A'+n-row;
```

```

        while (col<=row){
            cout<<ch<<" ";
            ch+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

 Writing pattern19.cpp

```

!g++ pattern19.cpp -o pattern19
!./pattern19

```

 5
E
D E
C D E
B C D E
A B C D E

%%writefile pattern20.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        int col=1;
        while (col<=row){
            cout<<char('A'+row+col-2)<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

 Writing pattern20.cpp

```

!g++ pattern20.cpp -o pattern20
!./pattern20

```

 5
A
B C
C D E
D E F G
E F G H I

▼ Triangle Numeric Printing

%%writefile pattern24.cpp

```

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        // space print
        int space = row-1;
        while (space){
            cout<<" "<<" ";
            space-=1;
        }
        // * print
        int col=1;
        while (col<=n-row+1){
            cout<<row<<" ";
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

```
}  
}
```

➦ Writing pattern24.cpp

```
!g++ pattern24.cpp -o pattern24  
!./pattern24
```

➦ 5

```
 1 1 1 1 1  
 2 2 2 2  
 3 3 3  
 4 4  
 5
```

%%writefile pattern25.cpp

```
#include<iostream>  
using namespace std;  
int main(){  
    int n;  
    cin>> n;  
    int row=1;  
    while (row<=n){  
        // space print  
        int space = n-row;  
        while (space){  
            cout<<" "<<" ";  
            space-=1;  
        }  
        // * print  
        int col=1;  
        while (col<=row){  
            cout<<row<<" ";  
            col+=1;  
        }  
        cout<<endl;  
        row+=1;  
    }  
}
```

➦ Writing pattern25.cpp

```
!g++ pattern25.cpp -o pattern25  
!./pattern25
```

➦ 5

```
      1  
     2 2  
    3 3 3  
   4 4 4 4  
  5 5 5 5 5
```

%%writefile pattern26.cpp

```
#include<iostream>  
using namespace std;  
int main(){  
    int n;  
    cin>> n;  
    int row=1;  
    while (row<=n){  
        // space print  
        int space = row-1;  
        while (space){  
            cout<<" "<<" ";  
            space-=1;  
        }  
        // * print  
        int col=row;  
        while (col<=n){  
            cout<<col<<" ";  
            col+=1;  
        }  
        cout<<endl;  
        row+=1;  
    }  
}
```

➦ Writing pattern26.cpp

```
!g++ pattern26.cpp -o pattern26
!./pattern26
```

```
5
1 2 3 4 5
  2 3 4 5
    3 4 5
      4 5
        5
```

```
%%writefile pattern27.cpp
```

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    int count = 1;
    while (row<=n){
        // space print
        int space = n-row;
        while (space){
            cout<<" "<<" ";
            space-=1;
        }
        // * print
        int col=1;
        while (col<=row){
            cout<<count<<" ";
            count+=1;
            col+=1;
        }
        cout<<endl;
        row+=1;
    }
}
```

```
Writing pattern27.cpp
```

```
!g++ pattern27.cpp -o pattern27
!./pattern27
```

```
5
      1
     2 3
    4 5 6
   7 8 9 10
  11 12 13 14 15
```

✓ Pascal Triangle Printing

```
%%writefile pattern28.cpp
```

```
#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        // space print
        int space = n-row;
        while (space){
            cout<<" "<<" ";
            space-=1;
        }
        // print 2nd triangle
        int col=1;
        while (col<=row){
            cout<<col<<" ";
            col+=1;
        }
        // print 3rd triangle
        int col1=row-1;
        while (col1){
            cout<<col1<<" ";
            col1-=1;
        }
    }
}
```

```

    }
    cout<<endl;
    row+=1;
}
}

```

↔ Overwriting pattern28.cpp

```

!g++ pattern28.cpp -o pattern28
!./pattern28

```

↔ 5

```

      1
     1 2 1
    1 2 3 2 1
   1 2 3 4 3 2 1
  1 2 3 4 5 4 3 2 1

```

%%writefile pattern29.cpp

```

/*
1 2 3 4 5 5 4 3 2 1
1 2 3 4 * * 3 4 2 1
1 2 3 * * * 3 2 1
1 2 * * * * 2 1
1 * * * * * 1
*/

#include<iostream>
using namespace std;
int main(){
    int n;
    cin>> n;
    int row=1;
    while (row<=n){
        // 1st triangle
        int col=1;
        while (col<=n-row+1){
            cout<<col<<" ";
            col+=1;
        }
        // 2nd triangle
        int col1=2;
        while (col1<=row){
            cout<<"*"<<" ";
            col1+=1;
        }
        // 3rd triangle
        int col2=row-1;
        while (col2){
            cout<<"*"<<" ";
            col2-=1;
        }
        // 4th triangle
        int col3=n-row+1;
        while (col3){
            cout<<col3<<" ";
            col3-=1;
        }
        cout<<endl;
        row+=1;
    }
}

```

↔ Overwriting pattern29.cpp

```

!g++ pattern29.cpp -o pattern29
!./pattern29

```

↔ 5

```

1 2 3 4 5 5 4 3 2 1
1 2 3 4 * * 4 3 2 1
1 2 3 * * * 3 2 1
1 2 * * * * 2 1
1 * * * * * 1

```

